Problem Set 2

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The Model, Pen and Paper

Question 1.1: Optimality Conditions

(a) Bellman equation and state variables

The household maximizes $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$ subject to $c_t + a_{t+1} = z_t + Ra_t$ with $R = 1 + r$. The budget constraint makes $c = z + Ra - a'$ a residual once tomorrow's assets are chosen. Hence,

$$v(a, z) = \max_{a' \in \mathcal{A}(a,z)} \{u(z + Ra - a') + \beta \mathbb{E}[v(a', z') \mid z]\} \tag{1}$$

(b) First-order condition

For an interior optimal choice, differentiation of the Bellman objective with respect to a' yields

$$\begin{aligned} 0 &= u_c(z + Ra - a')(-1) + \beta \mathbb{E}[v_a(a', z') \mid z] && (2)\\ &= -u_c(c) + \beta \mathbb{E}[v_a(a', z') \mid z]. && (3) \end{aligned}$$

Therefore,

$$u_c(c) = \beta \mathbb{E}[v_a(a', z') \mid z] \tag{4}$$

Saving one additional unit costs one unit of current consumption and raises the expected marginal continuation value of assets.

(c) Envelope condition, derived from the policy

Write the optimal policy as $a' = g(a, z)$ and insert it in the Bellman equation:

$$v(a, z) = u(z + Ra - g(a, z)) + \beta \mathbb{E}[v(g(a, z), z') \mid z]. \tag{5}$$

Differentiate with respect to current assets, holding current income fixed:

$$\begin{aligned} v_a(a, z) &= u_c(c)[R - g_a(a, z)] + \beta \mathbb{E}[v_a(g(a, z), z') \mid z] g_a(a, z) && (6)\\ &= R u_c(c) + g_a(a, z) \underbrace{[-u_c(c) + \beta \mathbb{E}[v_a(a', z') \mid z]]}_{0 \text{ by the FOC}} && (7) \end{aligned}$$

Thus

$$v_a(a, z) = R u_c(c) \tag{8}$$

An extra unit of current assets raises today's resources by R and is worth R times the marginal utility of consumption.

(d) Euler equation

Tomorrow's envelope condition is $v_a(a', z') = Ru_c(c')$, where $c' = z' + Ra' - g(a', z')$. Substituting it into the first-order condition from (b):

$$u_c(c) = \beta R \mathbb{E}[u_c(c') \mid z]. \tag{9}$$

This equates the cost of consuming one more unit now with the discounted expected return from transferring that unit to the next period. At a binding constraint, the condition is an inequality instead.

0.1 Question 1.2: The Analytical Household of Acharya and Dogra

(a) Euler equation, savings, and tomorrow's consumption

Set

$$u(c) = -\frac{1}{\gamma}e^{-\gamma c}, \quad u_c(c) = e^{-\gamma c}, \quad z = \bar{z} + e, \quad e' = \rho e + \sigma_\varepsilon \varepsilon', \quad \varepsilon' \sim N(0,1). \tag{10}$$

Writing cash on hand as $x = z + Ra$, the Euler equation becomes

$$e^{-\gamma c} = \beta R \mathbb{E}[e^{-\gamma c'} \mid x, e] \tag{11}$$

Conjecture $c = \mu x + \nu e + f$. The budget implies

$$a' = x - c = (1 - \mu)x - \nu e - f \tag{12}$$

Next-period cash on hand is

$$\begin{aligned} x' &= \bar{z} + e' + Ra' && (13)\\ &= R(1-\mu)x + (\rho - R\nu)e + \bar{z} - Rf + \sigma_\varepsilon \varepsilon'. && (14) \end{aligned}$$

Evaluating the same affine consumption rule tomorrow and collecting terms,

$$\begin{aligned} c' = \mu R(1-\mu)x + [\mu(\rho - R\nu) + \nu\rho]e \\ + \mu(\bar{z} - Rf) + f + (\mu + \nu)\sigma_\varepsilon \varepsilon'. \end{aligned} \tag{15}$$

Conditionally on (x, e), consumption is normal, with

$$\begin{aligned} m(x,e) &= \mu R(1-\mu)x + [\mu(\rho - R\nu) + \nu\rho]e + \mu(\bar{z} - Rf) + f, && (16)\\ s^2 &= (\mu + \nu)^2 \sigma_\varepsilon^2. && (17) \end{aligned}$$

(b) Compute the conditional expectation and match coefficients

For normal Y, $\mathbb{E}[e^Y] = \exp\{\mathbb{E}[Y] + \mathrm{Var}(Y)/2\}$. Taking $Y = -\gamma c'$ therefore yields

$$\mathbb{E}[e^{-\gamma c'} \mid x, e] = \exp\left[-\gamma m(x,e) + \frac{\gamma^2 s^2}{2}\right]. \tag{18}$$

Taking logs of the Euler equation and dividing by $-\gamma$ gives

$$c = m(x,e) - \frac{\ln(\beta R)}{\gamma} - \frac{\gamma s^2}{2} \tag{19}$$

Substitute $c = \mu x + \nu e + f$, $m(x, e)$ and s^2. Matching coefficients on x, on e, and the constants separately gives

$$\mu = \mu R(1 - \mu), \tag{20}$$
$$\nu = \mu(\rho - R\nu) + \rho\nu, \tag{21}$$
$$0 = \mu\bar{z} - \mu R f - \frac{\ln(\beta R)}{\gamma} - \frac{\gamma(\mu + \nu)^2\sigma_\varepsilon^2}{2} \tag{22}$$

For the nondegenerate policy with $\mu > 0$ and $r > 0$, the first relation gives $1 = R(1 - \mu)$, so $\mu = r/R$. The second gives $\nu[1 - \rho + \mu R] = \mu\rho$, and because $\mu R = r$ we have $\nu = \rho\mu/(R - \rho)$. Solving the constant equation for f gives

$$\begin{aligned} \mu &= \frac{r}{R}, \\ \nu &= \frac{\rho\mu}{R - \rho}, \\ f &= \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma(\mu + \nu)^2\sigma_\varepsilon^2}{2r} \end{aligned} \tag{23}$$

The last term reduces current consumption due to precautionary saving.

(c) Interpretation of μ

At fixed income deviation, $\partial c/\partial x = \mu = r/R$ is the marginal propensity to consume out of cash on hand. An additional unit of resources can finance an extra constant consumption stream d satisfying

$$1 = d\sum_{j=0}^{\infty} R^{-j} = \frac{d}{1 - R^{-1}} \quad \Longrightarrow \quad d = \frac{r}{R} = \mu. \tag{24}$$

Thus households consume the annuity value and save the remainder to fund later consumption.

(d) MPC out of an income innovation

A one-unit unexpected income innovation increases both cash on hand x and the income deviation e by one unit (with predetermined assets fixed). Therefore the marginal consumption response is $\mu + \nu$. Substitution gives

$$\text{MPC} = \mu + \nu = \frac{r}{R}\left(1 + \frac{\rho}{R - \rho}\right) = \frac{r}{R - \rho} \tag{25}$$

The derivative with respect to persistence is

$$\frac{\partial \text{MPC}}{\partial \rho} = \frac{r}{(R - \rho)^2} > 0. \tag{26}$$

When $\rho \to 0$, the innovation is transitory and MPC $\to r/R$: only its annuity value is consumed. When $\rho \to 1$, MPC $\to r/(R - 1) = 1$: a nearly permanent increase in labor income supports an almost one-for-one consumption increase.

(e) Optimal policies for i.i.d. income

Set $\rho = 0$, so $\nu = 0$ and $\sigma = \sigma_\varepsilon$. The optimal consumption and savings policies become

$$c(x) = \frac{r}{R}x + \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma r\sigma_\varepsilon^2}{2R^2}, \tag{27}$$
$$a'(x) = \frac{x - \bar{z}}{R} + \frac{\ln(\beta R)}{\gamma r} + \frac{\gamma r\sigma_\varepsilon^2}{2R^2} \tag{28}$$

Equivalently, $a'(a,z) = a + (z - \bar{z})/R + \ln(\beta R)/(\gamma r) + \gamma r \sigma_\varepsilon^2/(2R^2)$.

The Model on a Computer: The CRRA Household

Question 3.1: From one model to the other

(a) Utility functions

In the CARA model,

$$u(c) = -\frac{e^{-\gamma c}}{\gamma}, \qquad u_c(c) = e^{-\gamma c}, \qquad u_c^{-1}(m) = -\frac{\ln m}{\gamma} \tag{29}$$

CARA utility and marginal utility are defined for all $c \in \mathbb{R}$, and inverse marginal utility requires $m > 0$. In the CRRA model,

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \qquad u_c(c) = c^{-\gamma}, \qquad u_c^{-1}(m) = m^{-1/\gamma} \tag{30}$$

For $\gamma = 2$,

$$u(c) = -\frac{1}{c}, \quad u_c(c) = \frac{1}{c^2}, \quad u_c^{-1}(m) = \frac{1}{\sqrt{m}} \tag{31}$$

CRRA utility and marginal utility require $c > 0$; inverse marginal utility requires $m > 0$.

(b) Income process

CARA assumes normal income in levels,

$$z = \bar{z} + e, \quad e' = \rho e + \sigma_\varepsilon \varepsilon' \tag{32}$$

Rouwenhorst discretizes e and income in levels is obtained by adding $\bar{z}$. In CRRA, log income follows

$$\ln z' = \rho \ln z + \sigma_\varepsilon \varepsilon', \qquad \varepsilon' \sim N(0,1) \tag{33}$$

The stationary standard deviation of $y = \ln z$ is

$$\sigma_y = \frac{0.2}{\sqrt{1 - 0.9^2}} \approx 0.458831 \tag{34}$$

Income in levels is then obtained by exponentiating the discretized log-income states:

$$z_m = \exp(y_m) \tag{35}$$

Therefore, income has a long right tail: a few very high incomes and many moderate or low incomes. High-income shocks can be much larger than low-income shocks in absolute terms

(c) Asset grid

The CARA model uses 400 equally spaced points on $[-15, 25]$, allowing negative assets. The CRRA household faces $a' \geq 0$, so its lower grid bound is zero. The upper bound is artificial and must cover the economically relevant savings decisions and the stationary wealth distribution. We use 200 quadratic points on $[0, 100]$. This grid is dense near zero, where the borrowing constraint can bind and the policy may have a kink. The artificial upper boundary should be checked for truncation effects.

(d) Borrowing constraint and consumption bounds

At (a_i, z_m) the budget is $c_{im} + a'_{im} = x_{im}$ with $x_{im} = z_m + Ra_i$. Since $a' \geq 0$ and CRRA requires positive consumption,

$$0 < c_{im} \leq x_{im} \tag{36}$$

With the additional numerical upper bound $a' \leq 100$, the code enforces

$$c_{im}^{\min} = \max\{x_{im} - 100, 10^{-10}\}, \qquad c_{im}^{\max} = x_{im} \tag{37}$$

The numerical floor avoids evaluating marginal utility at zero. When borrowing is constrained, $a' = 0$, so $c = x$.